Inside the Earth | Poster presentation script

Suggested delivery: about 6–7 minutes including pauses to point to the tables and figures. This is a rehearsal estimate, not a course rule. Follow the five columns from left to right.

Opening | five columns, one catalogue

Hello. We are the DSC5002 project team at City University of Hong Kong. Inside the Earth studies earthquake patterns through space, time, depth and magnitude. The poster reads from left to right through five columns. Its tables give precise comparisons, while five selected figures show geography, variation and structure. We describe a fixed catalogue; we do not predict future earthquakes.

Column 1 | study design

The first column explains the motivation and common design. A map can show a cluster, but it cannot establish whether that pattern comes from geography, a large sequence or a change in reporting. Table 1 defines our sample: 46,463 unique USGS earthquakes from 2000 to 2025, all with preferred magnitude at least five. All records have coordinates, magnitude and depth. We retain four negative reported depths and exclude non-earthquake event types. The world map then supplies the spatial setting. Completeness of the downloaded files is not proof of complete detection.

Column 2 | space and reference

Figure 2 and Table 2 ask: concentrated relative to what? We compare observed boundary distances with 65,536 equal-area reference locations over the whole globe, including oceans. A second reference preserves the observed latitude distribution while varying longitude. Within one hundred kilometres, 70.15 percent of events are included, compared with 9.42 percent of equal-area points and 10.50 percent of latitude-controlled points. The observed-to-area share ratio is 7.45. These are geographic comparisons, not a realistic earthquake-occurrence model. The range across four reference designs measures numerical sensitivity; it is not a confidence interval or a hazard multiplier.

Column 3 | time and comparability

The third column places the annual trend beside Table 3, which documents measurement comparability. Counts peak in 2011. The later thirteen-year period has a 6.63 percent lower record rate at magnitude five and a 15.42 percent lower rate at magnitude six. Yet the whole-period M5 Sen slope is positive, so the chosen summary matters. Mw-family share changes from 73.45 to 55.05 percent, while horizontal-error availability rises sharply. Regional contrasts also differ in direction. Removing fixed windows around the twenty-eight largest events reduces the global M5 contrast to minus 3.37 percent. This is a sensitivity check, not declustering.

Column 4 | depth and associations

Table 4 shows why medians are insufficient. Japan and Kuril and Tonga, Kermadec and Fiji have the same thirty-three-kilometre median, yet deep-event shares are 5.01 and 11.76 percent. Figure 4 pairs local route maps with sections. The star is the mapped trench origin; A to B gives increasing section distance, and horizontal and vertical kilometres use equal scales. The selected Tonga section contains 186 deep events out of 640; northern Tohoku contains 21 out of 351. The routes were chosen after exploration. Regional depth-distance associations support a structured descriptive pattern, but do not establish a causal mechanism.

Column 5 | magnitude and event sequences

The last column uses two compact tables. Table 5 compares event-count shares with an illustrative magnitude-based proxy. The twenty-eight magnitude-eight-or-larger events are only 0.06 percent of records but contribute 71.21 percent of the summed proxy. Because preferred magnitude scales are mixed, this is not measured energy or damage. Table 6 then compares three events using the same five-hundred-kilometre radius and thirty-day before and after windows, excluding each anchor. Figure 5 normalizes the post-window counts to compare timing. About sixty to seventy-one percent appear in the first three days. Membership in a time-and-distance window does not prove that an event is a causal aftershock.

Closing | what the evidence supports

The combined evidence shows concentration near mapped boundaries, contrasting regional depth tails and clustered selected sequences. Our interpretation depends on reference geography, catalogue composition and window definition. The contribution is to make those choices visible through explanatory text, inspectable tables and focused figures. Thank you for your attention. We would be happy to discuss the reference design or the supplementary sensitivity results.

Discussion prompts

Why not use every plot? The poster selects five figures that add information to the narrative; precise comparisons are shown in six tables. All fourteen analytical figures remain in the report.

Why not fit a predictive b-value? An M5 extraction threshold does not establish completeness, and preferred magnitude scales are mixed.

Are the references a statistical null model? No. They control geometric area and latitude, not the tectonic processes governing earthquake occurrence.

Presentation reference

CityU Language Centre: poster script sample. The opening, signposting, methods, findings and closing structure follow the sample; the research content and wording are original.